

11-14 Reflection

Prerequisites

Students should have already completed the **Light Sources and Rays** sheets. Completing the **Energy** sheets would be helpful, but is not strictly necessary.

Maths: Students should recall that angles in a right angle sum to 90° and be able to subtract one angle away from this to find the remaining angle. They should also be comfortable with using division to find equal parts.

The Main Points

- The technical term for when light bounces is **reflection**
- Light always travels in **straight lines**
- **Reflection** can be **specular** (light only leaves the surface in one direction) or **diffuse** (light is scattered upon reflection)
- In ray diagrams, it is often useful to draw a line at a **right angle** to the surface called a **normal**
- The **angles of incidence and reflection** are measured from the normal. **This is introduced by discovery in Class Q5.**
- The angle of incidence is equal to the angle of reflection
- Protractors need to be lined up with a valid zero line and centered and the relevant vertex.

Teacher Quarter Briefing

- Introduction: Video is being prepared - coming soon!
- Practice: https://isaacscience.org/question_decks#itsp_teach_reflection
- Review: Video is being prepared - coming soon!
- If you want to go further: If you want to go further, please see the reflection materials in Essential GCSE Physics on [Plane Mirrors](#), [Concave Mirrors](#) and [Convex Mirrors](#).

Class Question Notes

The worksheet can be printed either in full, or in cloze text form (where the red text is missing, and students can complete these blank spaces after class discussion). The online version of the notes requires the appropriate text to be dragged to the right place in the sentences.

[Shallow learning gradient online assignment](#) - q1,2,3,4,5,6,7,8,10

[Steeper learning gradient online assignment](#) - q1,2,3,4,5,6,7,9,10,11

1. This question has students distinguish between a ray diagram showing specular reflection and one showing diffuse reflection, reinforcing the definition that diffuse involves scattering, specular involves maintaining a single direction.
2. This question asks students to identify whether different concrete examples demonstrate specular or diffuse reflection. These can be reasoned through either by considering the roughness of a surface, and hence scattered reflection, or phenomenologically considering whether they would see an image reflected. Correct answers:
Mirror → Specular (hence we see a clean reflection)
Painted Wall → Diffuse (hence a matt effect - no visible 'reflection' [image])
Window → Specular (hence we see an image)
Clothes → Diffuse (uneven fibres scatter light - no image)
3. Asks students to explicitly consider whether diffuse or specular reflection is needed for an image to be maintained after reflection. Correct answer: specular, otherwise the relative position/direction of rays changes, changing the shape they make.
4. A fill in the gaps question that has students complete an explanation of the fact that brightness can be (approximately) maintained in specular reflection as all reflected light maintains a spatially dense reflected ray, while diffuse reflection sends rays in many directions, so each ray is less bright (Conservation of energy link).
5. Students identify different parts of a reflection ray diagram. **Students uncover the definitions of incident and reflected rays by discovery.** We leave it at teachers' discretion whether to explicitly supply said definitions in advance, depending on the needs of the class.
6. Students are given various angles on diagrams showing incident rays and need to carefully consider whether they are the angle of incidence, or whether they need to **use the fact that the normal is at 90° to work it out** using a 1 step difference calculation. **Some students will need explicit maths support.**
7. Part (a) asks students to identify correct protractor usage. Correct answers: B and C. Part (b) has them measure the angle: they will need to divide by 3 to share 90° into 3 equal parts. Correct answer: 30°. Part (c) has them explicitly state that the protractor needs to be centred on the relevant vertex (the point where the ray hits the surface)
8. Students measure angles of incidence and reflection 6 times, hopefully noticing that they are always the same. The online version is implemented similarly to q7. Angles are easy increments, should be measurable, but you may want to **teach that extending the rays makes protractor measurement easier**
9. Has students reflect on their answers to question 8, hopefully drawing out that $i = r$.
10. Students are presented with a ray diagram where $i = 0^\circ$ and asked (a) to identify this, (b) predict the angle of reflection and (c) draw the reflected ray. An analogy with a bouncing ball may be helpful.
11. Students (a) measure the angle of incidence of a ray (**65°**), (b) Draw reflected rays, (c) Extend said rays through the reflecting surface (d) notice that **all extended reflected rays meet at a single point vertically below the point of emission, at the same distance from the mirror.** This is designed to demonstrate why it looks to our eyes why it looks like the rays came from behind the mirror. Note that the entire image one sees in the mirror comes from extending this logic to consider every point in space in front of the mirror.

Homework Question Notes

These questions have a very similar form to the questions in the class task, so students can refer back to their earlier answers to help

[Shallow learning gradient online assignment](#) - q1,2,4,5,6,7,8

[Steeper learning gradient online assignment](#) - q1,2,3,4,7, 8,9,10,11

1. Similar to Class Q1, has students identify ray diagrams for specular/diffuse reflection. Correct answers: (a) A, is specular (b) B is diffuse
2. Similar to Class Q2, students consider concrete examples of specular/diffuse reflection. Correct answers: Polished spoon → Specular (subtle: distortion comes from curved surface, not scattering), Paper → diffuse, Pavement → diffuse, Still puddle → specular
3. Students consider smudging a surface to make it less optically smooth and identifying this with **diffuse reflection (answer)**
4. Fill in the gaps having students consider concrete examples of uses of diffuse vs specular reflection in acoustic engineering. A lecture theatre may use rough surfaces to induce diffuse reflection, preventing echoes. A place of worship is often designed to use smooth surfaces to induce echoes in specific directions, helping carry sound to worshippers.
5. Similar to Class Q5, students identify features of a ray diagram. The example is a basketball hitting a backboard (taking either a high velocity or low distance approximation to justify straight trajectory), reinforcing the concretising mental schema connecting reflection to bouncing.
6. Similar to Class Q6, students identify whether a given angle directly states the angle of incidence or whether it is a difference between said angle and the normal (90°), **requiring a simple calculation**.
7. Similar to Class Q7, students identify **incorrect** protractor usage. Correct answers (a) C (witty/mathematical students may argue they can apply the methods from Q6... great!) (b) 60° **students will need to be able to use $\div$ to find parts and $+/ \times$ to combine them** (c) the point where the ray meets the surface.
8. Similar to Class Q8, students measure angles of incidence/reflection.
9. This question is designed to encourage students to reflect on the symmetries of reflection. A scenario is presented where only the shape of the trajectory (not the direction of travel) is shown in 3 different videos: an original reflection, a time reversal and the original with incident and reflected rays swapped. Students are asked to explain whether and why they can/cannot be distinguished. The correct answer is that they cannot because the angle of incidence always equals the angle of reflection and all small sections of each ray are indistinguishable, so the trajectories look the same in all cases.
10. Students are asked to compare the angles of reflection in two cases that are identical except that one surface absorbs some of the light. The answer is that the angles of reflection are the same because they are determined solely by the angle of incidence.
11. Mirrors are set up perpendicular to a table in a closed square shape and a ray of light is sent towards one with an angle of incidence of 45° . Students are asked what shapes the ray could trace as it is reflected. The correct answer is rectangles (including the special case of a square). Follow up questions may include asking what determines the length:width ratio, whether a line can be made, and whether similar answers exist for other mirror arrangements.

Extension questions from Essential GCSE Physics:

https://isaacscience.org/question_decks#gcse_phys_ch5_39